

VG12 – SPIN AS A PERMISSIVITY FILTER: FROM PAULI TO VG

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Abstract

In standard physics, spin is an intrinsic property of particles, with no clear physical mechanism explaining why fermions and bosons behave differently. VG12 proposes a radical reinterpretation: **spin is not a static property, but a dynamic filter** that determines the permissivity of coupling between nodes. Thus, quantum statistics (Fermi-Dirac vs. Bose-Einstein) and the Pauli exclusion principle become **emergent consequences** of network dynamics, not externally imposed rules. This view offers a unified explanation for the behavior of fermions and bosons, and opens the way for a deeper understanding of matter and forces.

Keywords: spin, permissivity filter, fermions, bosons, Pauli principle, quantum statistics, VG.

1. INTRODUCTION – SPIN AS A MYSTERY OF STANDARD PHYSICS

In standard physics, spin is an intrinsic property of particles, with two major consequences:

- **Fermions** (half-integer spin) are subject to the **Pauli exclusion principle**: two identical particles cannot occupy the same quantum state.
- **Bosons** (integer spin) are not subject to this restriction and can form collective states (laser, Bose-Einstein condensation).

Why does this difference exist? Standard physics postulates it, but does not explain it. VG12 offers an emergent explanation: **spin is a dynamic filter** that determines the permissivity of coupling in the network.

2. THE COUPLING HAMILTONIAN – FROM INITIAL TO COMPLETE FORM

In VG, the coupling Hamiltonian was initially defined as:

$$H_{\text{coupling}}^{\text{initial}} = \sum_{i < j} K_{ij} q_i q_j$$

This describes **pure vibrational coupling** between two nodes, independent of the orientation of their spins.

In VG12, we extend this formula to include **spin as a permissivity filter**:

$$H_{\text{coupling}}^{\text{complete}} = \sum_{i < j} K_{ij} \cdot (1 - \hat{S}_i \cdot \hat{S}_j) \cdot q_i q_j$$

Explanation of the additional term:

- $\hat{S}_i, \hat{S}_j$ are the spin operators of nodes i and j .
- $\hat{S}_i \cdot \hat{S}_j$ measures the **alignment** of the spins.
- $1 - \hat{S}_i \cdot \hat{S}_j$ acts as a **filter**:
 - If the spins are **aligned** ($\hat{S}_i \cdot \hat{S}_j > 0$), the term is small $\rightarrow$ coupling **permitted** (bosonic behavior).
 - If the spins are **anti-aligned** ($\hat{S}_i \cdot \hat{S}_j < 0$), the term is large $\rightarrow$ coupling **reduced** (fermionic behavior).

This completion transforms spin from a static property into an **active coupling regulation mechanism**, automatically generating fermionic and bosonic behaviors.

3. FERMIONS AND BOSONS – EMERGENCE OF QUANTUM STATISTICS

- **Fermions** (half-integer spin) have a **very restrictive filter**. Once a vibrational state is occupied, another identical one is **blocked**. This is the **reduced permissivity** that gives matter its structure.

- **Bosons** (integer spin) have a **very permissive filter**. They allow many identical bosons to occupy the same vibrational state. This is the **high permissivity** that leads to collective phenomena (laser, Bose-Einstein condensation).

Quantum statistics (Fermi-Dirac, Bose-Einstein) become **emergent properties** of network dynamics, not external postulates.

4. THE PAULI EXCLUSION PRINCIPLE – A CONSEQUENCE OF NETWORK DYNAMICS

The Pauli principle becomes a **natural consequence** of how a node's spin filter limits the occupation of neighboring vibrational states. If a node is already occupied by a fermion, another identical fermion **cannot** enter the same state, because the spin filters are incompatible.

This is no longer an externally imposed rule, but a **property of the network** – just as two balls of the same color cannot occupy the same space at the same time.

5. CONSEQUENCES FOR VG

1. **Quantum statistics** – become emergent properties of network dynamics, not external postulates.

2. **Stability of matter** – the Pauli principle, which ensures the stability of atoms, is a consequence of the network.

3. **Collective phenomena** – lasers, Bose-Einstein condensation and superconductivity are expressions of the high permissivity of bosons.

4. **A new understanding of spin** – no longer a mysterious property, but an **active coupling regulation mechanism**.

6. PREDICTIONS AND OPEN QUESTIONS

- **Correlations between spin and coupling** – can we test whether K_{ij} depends on spin alignment in controlled quantum systems?
- **Modifications of quantum statistics** – under extreme coupling conditions, statistics might be modified (fermion-boson transitions).
- **Spin as a dynamic variable** – can we model spin as a time-evolving variable, not a constant?

7. CONCLUSION

VG12 proposes a radical reinterpretation of spin: **a dynamic filter** that determines the permissivity of coupling in the network. From this perspective, quantum statistics and the Pauli principle become **emergent consequences** of network dynamics, not externally imposed rules. This is a **unification** of two seemingly separate domains – quantum statistics and network dynamics – and a **confirmation** that VG offers a coherent framework for understanding reality at the fundamental level.